Rishi Sappidi

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Education

University of Washington Seattle, WA
B.S. Electrical and Computer Engineering, GPA: 3.54/4.00 *June 2027*
- Coursework: Computer Architecture I/II, Digital Design, VLSI I/II, Semiconductor Devices

Publication

J. Zhang, M. Nie, J. Wang, W. Li, **R. Sappidi**, S. Chitnis, A. Li. "CacheFlex: Direct Software-Managed Access to Higher-Level Cache for Scalable Vector Support." Accepted to IEEE/ACM MICRO, 2026.

Experience

PN Computer Engineering Lab, University of Washington Nov. 2025 – Present
Undergraduate Researcher – Computer Architecture & Memory Systems
- Lead the coherent multicore extension of CacheFlex, exposing private L2 cache ways as software-managed scratchpad memory; contributed to the CacheFlex paper accepted at MICRO 2026.
- Extended gem5 Ruby/SLICC MESI protocols with scratchpad requests, transient states, invalidation behavior, cache-line migration, and cache/scratchpad handoff; developed Rumur models for coherence invariants.
- Optimized and numerically validated Qwen2.5-0.5B-shaped prefill kernels in eight-core gem5: private-scratchpad speedups of 1.36× for SwiGLU FFN and 1.30–1.31× for QKV/output projections at 1,024 tokens.
- Achieved 1.04–1.11× speedups for causal grouped-query FlashAttention at 1,024–4,096 tokens; compared against independently tuned cache baselines using bounded searches and included packing/staging costs in kernel evaluations.

Silicon Systems Research Lab, University of Washington Jun. 2026 – Present
Undergraduate Researcher – Accelerator Architecture & RTL
- Co-designed a hierarchical Ising accelerator targeting 1,048,576-spin capacity with 32-spin cores, banked state SRAM, and a FlooNoC mesh; memory-side engines at each chiplet's DRAM interface read each coupling block once, compute both directions, and ship partial sums to endpoint cores.
- Built a C++ exact-event model integrating Ramulator2 HBM4/GDDR6 timing and cycle-level network contention; verified it cycle-for-cycle against SystemVerilog RTL in VCS on 24 differential cases up to a 65,536-spin dense case.
- Implemented a MAC-matched flat weight-stationary baseline; the symmetric single-fetch design used 1.65–1.74× fewer simulated iteration cycles on dense workloads (64K–512K spins) and 1.53–1.62× fewer on toroidal graphs through 1M spins.
- Diagnosed scaling limits from model counters, including a remote partial-sum port bound on dense runs and flush serialization on unstructured sparse graphs beyond 512K spins; identified wider-port and early-flush redesigns.

Sensors, Energy, and Automation Lab (SEAL), UW Jun. 2025 – Nov. 2025
Embedded Team Lead – SpectraSense
- Led a six-person team developing a handheld pesticide-screening prototype; designed a four-layer STM32H7/spectrometer PCB and implemented PARAFAC/N-PLS edge-classification methods. Collegiate Inventors Competition finalist.

Monarch Tractor Jun. 2024 – Sept. 2024
Intern Developer – Computer Vision/Graphics
- Built an Open3D point-cloud segmentation tool with 10+ annotation features, enabling on-site dataset labeling and reducing annotation outsourcing costs by 10%.

Selected Hardware Projects

Pipelined 64-bit ARM CPU | *SystemVerilog, ModelSim*
- Designed a five-stage processor with forwarding, load-use hazard detection, and branch control; verified instruction execution and cycle-accurate pipeline behavior using directed assembly tests and waveform analysis.

GCD Accelerator ASIC Implementation | *SystemVerilog, Cadence*
- Integrated compute RTL, hard SRAM, and packetized NoC interfaces through synthesis and post-layout verification; achieved 220.6 MHz, 6,554 µm² logic area, and 1.341 mW in the evaluated flow.

16-bit, 13-Entry Register File | *HSPICE, Cadence Virtuoso, FreePDK45*
- Designed latch storage, write decoding, and read mux logic for a full-custom 2R1W register file; achieved a DRC/LVS-clean 52×60 µm layout operating at 1.43 GHz post-layout, with 31.6 pJ read and 1.2 pJ write energy.

Technical Skills

Architecture: Memory Hierarchies, Cache Coherence, Scratchpad Memory, NoCs, Accelerator Architecture, Microarchitecture
Modeling & Analysis: gem5 (Ruby/SLICC), Ramulator2, Performance Modeling, Benchmarking, Traffic/Bottleneck Analysis
Languages: C/C++, Python, SLICC, SystemVerilog, Verilog, ARM/RISC-V Assembly, Tcl, Bash
Verification & Tools: Rumur, VCS, Verilator, ModelSim, Vivado, Quartus, Cadence Genus/Innovus/Tempus, Linux, Git